

BloodLink Functional Testing Form

Blood Inventory Monitoring and Decision Support System

This form documents functional test cases for BloodLink. Execute each case using the stated scenario, record the observed outcome in Actual Result, and note evidence, defects, or follow-up actions in Remarks.

Project: BloodLink	Testing Type: Functional Testing	Tester: _____	Date: _____
Build or Version: _____	Test Environment: _____	Result Codes: P Passed F Failed PEND Pending	Evidence Reference: _____

1. Authentication and Role Based Access

FR#	Description	Expected Result	Actual Result	Remarks
FR1.1	Log in with valid seeded credentials for each authorized role	System authenticates the user and opens the correct role dashboard.		
FR1.2	Log in using an incorrect password	System denies access and displays a login error message.		
FR1.3	Log in using an unregistered email address	System denies access and displays an account-not-found or login error message.		
FR1.4	Submit login with required fields blank	System identifies the required email and password fields.		
FR1.5	Open a protected dashboard without an authenticated session	System prevents unauthorized access to the protected dashboard.		
FR1.6	Log out from a portal	System ends the session and returns the user to the landing or login page.		
FR1.7	Verify role restricted modules	User sees only modules and actions assigned to their role.		

2. Donor Management Unit

FR#	Description	Expected Result	Actual Result	Remarks
FR2.1	Register a donor with complete and valid required information	New donor record is saved and appears in donor management.		
FR2.2	Register a donor with incomplete required information	System prevents submission and identifies the missing fields.		
FR2.3	Search an existing donor by name or donor identifier	Relevant donor record is displayed.		
FR2.4	View an existing donor profile	Correct demographic, blood type, and donation information is displayed.		
FR2.5	Update a donor profile	Updated donor information is saved and displayed correctly.		

FR#	Description	Expected Result	Actual Result	Remarks
FR2.6	Record a donation with valid donor and collection details	Donation record is saved and linked to the correct donor.		
FR2.7	Record a donation with missing required information	System prevents submission and identifies missing fields.		
FR2.8	View donation records	Donation records display the correct donor, serial number, date, and status.		
FR2.9	Search or filter donation records	Only records matching the selected search or filter criteria are displayed.		

3. Serology and Transmissible Disease Screening

FR#	Description	Expected Result	Actual Result	Remarks
FR3.1	View donations awaiting laboratory results	Pending donations are displayed for serology processing.		
FR3.2	Open a pending donation from its serial number	Laboratory result form opens with the selected donation details.		
FR3.3	Encode complete non-reactive screening results	Lab result is saved and the donation is marked Accepted or Cleared according to the outcome.		
FR3.4	Encode reactive or indeterminate test results	System records the result and applies the appropriate screening outcome or quarantine status.		
FR3.5	Submit a laboratory result with missing required fields	System prevents submission and identifies required fields.		
FR3.6	Search or filter laboratory results	Only results matching the selected serial number or result filter are displayed.		
FR3.7	View an encoded laboratory result	Saved blood type, TTI results, outcome, and remarks are displayed correctly.		

4. Blood Component Production and Inventory

FR#	Description	Expected Result	Actual Result	Remarks
FR4.1	View serology-cleared donations awaiting component processing	Eligible donations are listed for processing.		
FR4.2	Open component processing using a cleared donation serial number	Processing form opens with the selected serial number.		
FR4.3	Record one or more blood components with valid details	Component units are saved and linked to the correct donation.		
FR4.4	Attempt to process a donation without a laboratory result	System prevents processing and informs the user that laboratory results are required.		
FR4.5	Attempt to process a deferred or non-accepted donation	System prevents component processing for the ineligible donation.		

FR#	Description	Expected Result	Actual Result	Remarks
FR4.6	Enter a component volume outside the acceptable range	System flags the entry and applies the required safety handling or validation.		
FR4.7	View blood component inventory ledger	Recorded units display the correct serial number, blood type, component, and status.		
FR4.8	Search or filter inventory by component or status	Only matching inventory units are displayed.		
FR4.9	View units approaching expiration	Expiring units are correctly identified in the inventory indicators.		

5. Hospital Blood Requests and Issuance

FR#	Description	Expected Result	Actual Result	Remarks
FR5.1	Hospital user creates a blood request with complete valid information	Request is submitted, assigned a reference number, and appears in the hospital request list.		
FR5.2	Hospital user submits a request with required fields missing	System prevents submission and identifies the missing information.		
FR5.3	Hospital user views submitted blood requests	Only the user hospital's requests and their current statuses are displayed.		
FR5.4	Issuance Personnel views incoming blood requests	Request queue displays submitted hospital requests with correct details and status.		
FR5.5	Verify a submitted blood request	Request status updates to Verified when stock and request details are validated.		
FR5.6	Reject an ineligible or incomplete request	System records the rejection and displays the updated request status.		
FR5.7	Process an approved blood request	System records the issuance workflow and updates request processing status.		
FR5.8	Approve blood release	Release status is recorded and issued units are reflected in the issuance record.		
FR5.9	Record a walk-in issuance	Walk-in issuance is saved with correct unit, recipient, and release details.		
FR5.10	View issuance audit log	Audit log displays completed issuance and release details correctly.		

6. Donor Recall and Notifications

FR#	Description	Expected Result	Actual Result	Remarks
FR6.1	View donors eligible for recall based on the donation interval	System identifies and lists eligible repeat donors.		
FR6.2	Exclude a donor not yet eligible for recall	Ineligible donor is not included in the eligible recall list.		
FR6.3	Generate a donor recall message	System creates a reminder message for the selected eligible donor.		
FR6.4	Send a recall reminder through PhilSMS	System sends the notification request and records the delivery or sending status.		
FR6.5	View donor recall history	Recall records show donor, message date, and notification status.		

7. Demand Forecasting and Equity Based Distribution

FR#	Description	Expected Result	Actual Result	Remarks
FR7.1	Generate a Multiple Linear Regression demand forecast using available historical distribution data	System produces forecast values by blood type for the selected forecast period.		
FR7.2	Filter demand forecast by blood type, hospital, or forecast period	Forecast display updates to the selected criteria.		
FR7.3	View demand versus inventory gap analysis	System displays predicted demand, available inventory, gap, and shortage or surplus indication.		
FR7.4	Generate equity based distribution recommendations	System recommends proportional component allocation using predicted demand and available inventory.		
FR7.5	View distribution recommendation details for a hospital	Recommended units, basis of allocation, and inventory context are displayed.		
FR7.6	Generate recommendations when available inventory is insufficient	System returns proportional or constrained recommendations without allocating more units than available.		

8. Administration and Audit

FR#	Description	Expected Result	Actual Result	Remarks
FR8.1	Super Admin views the user management list	System displays user accounts with correct role and account details.		
FR8.2	Create a user account with a valid assigned role	New user account is saved and assigned the selected role.		
FR8.3	Create a user account with duplicate email	System prevents duplicate account creation and displays a validation message.		

FR#	Description	Expected Result	Actual Result	Remarks
FR8.4	Update an existing user account	Updated user information and role are saved correctly.		
FR8.5	Deactivate or reactivate a user account	Account status changes and access follows the selected status.		
FR8.6	View audit activity or system records	Relevant activity records display the user, action, and timestamp correctly.		

Functional Testing Summary

Complete this summary after executing the individual functional test cases. Keep Pending separate from Failed until a final result is established.

Module	Passed	Failed	Pending	Total	Pass Rate
Authentication and Role Based Access				7	
Donor Management Unit				9	
Serology and Transmissible Disease Screening				7	
Blood Component Production and Inventory				9	
Hospital Blood Requests and Issuance				10	
Donor Recall and Notifications				5	
Demand Forecasting and Equity Based Distribution				6	
Administration and Audit				6	
TOTAL				59	

Suggested computation: Total Test Cases = Passed + Failed + Pending. Pass Rate = Passed divided by (Passed + Failed) multiplied by 100. Do not include Pending in the Pass Rate until it has been executed.